

Performance of Algorithms Submitted in the 2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge

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Background: The 2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge invited participants to develop artificial intelligence (AI) models capable of independently interpreting mammograms.

Purpose: To assess the performance of the submitted algorithms, explore the potential for improving performance by combining the best-performing AI algorithms, and investigate how performance was influenced by the demographic and clinical characteristics of the evaluation cohort.

Materials and Methods: A total of 1687 AI algorithms were submitted from November 2022 to February 2023. Of these, 1537 algorithms were assessed using an evaluation dataset from two sites—one in the United States and one in Australia. Cancer cases were identified at screening and confirmed with pathologic examination; noncancer cases were followed up for at least 1 year. Results for ensemble models of top algorithms were computed by recalling a case when any of the included algorithms indicated recall. Odds ratios (ORs) were used to investigate differences in AI performance when the dataset was stratified by clinical or demographic characteristics.

Results: The evaluation dataset consisted of 5415 women (median age, 59 years [IQR, 52–66 years]). Among the 1537 AI algorithms, the median recall rate, sensitivity, specificity, and positive predictive value (PPV) were 1.7%, 27.6%, 98.7%, and 36.9%, respectively. For the top-ranked algorithm, the recall rate, sensitivity, specificity, and PPV were 1.5%, 48.6%, 99.5%, and 64.6%, respectively. Ensemble models of the top 3 and top 10 algorithms had a sensitivity of 60.7% and 67.8%, respectively; the corresponding recall rates were 2.4% and 3.5%, and the corresponding specificities were 98.8% and 97.8%. Lower sensitivity was observed for the U.S. dataset than for the Australian dataset (top 3 ensemble model: 52.0% vs 68.1%; OR = 0.51; $P = .02$), and greater sensitivity was observed for invasive cancers than for noninvasive cancers (top 3 ensemble model: 68.0% vs 43.8%; OR = 2.73; $P = .001$).

Conclusion: The different AI algorithms identified different cancers during screening mammography, and ensemble models had increased sensitivity while maintaining low recall rates.

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Supplemental material is available for this article.

Breast cancer is the most commonly diagnosed cancer worldwide, with 2.26 million new cases and 685 000 deaths reported in 2020 alone (1). Routine mammographic screening has been proven to significantly reduce breast cancer mortality by facilitating early detection and treatment (2,3). However, currently, breast screening programs worldwide rely on highly skilled radiologists to interpret screening mammograms. With large screening populations, this is a costly and time-consuming endeavor, especially for countries with nationally organized programs where the standard is to double-read mammograms, as found in Europe and Australia (4). Furthermore, although screening programs are successful in detecting breast cancer, they are not flawless: Missed cancers still manifest as interval cancers with poor prognoses, and high recall rates can lead to further unnecessary patient workups (5,6).

Recent advances in artificial intelligence (AI) methodologies, particularly deep learning, provide opportunities to improve breast screening programs by incorporating AI models that can independently analyze mammograms (7,8). AI tools with performance comparable to that of human readers (9) could be integrated into the screening workflow, reducing radiologist workload by serving as a triaging tool, excluding normal examinations from reporting, or acting as an independent reader

in double-reading settings (10–14). These AI tools could also increase the cancer detection rate and reduce the recall rate. Alternatively, AI models could be used as decision support tools to improve reporting accuracy and efficiency (15,16).

In recent years, crowdsourced competitions have emerged as a means to engage the international scientific community in solving complex biomedical problems, promoting collaboration and driving improvements in health care efficiency and patient outcomes (13,17). RSNA has hosted several AI challenges to encourage the development of AI tools for radiology (18). In the 2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge, participants were invited to develop AI models capable of independently interpreting mammograms. The curated imaging dataset for this challenge and the work submitted by the contestants are expected to lay a valuable foundation for further rapid advances in breast imaging AI. The competition attracted 2146 competitors, forming 1687 teams, each submitting their own AI algorithm. The aims of this study were to assess the performance of the submitted algorithms, explore the potential for improving performance by combining the highest-ranked AI algorithms, and investigate how performance was influenced by patient characteristics in the evaluation cohort.

Abbreviations

AI = artificial intelligence, OR = odds ratio, PPV = positive predictive value

Summary

Algorithms submitted to the RSNA Screening Mammography Breast Cancer Detection AI Challenge detected different cancers; ensemble models had increased sensitivity while maintaining low recall rates; and sensitivity was higher for invasive versus noninvasive cancers.

Key Results

- Among 1537 AI algorithms evaluated in the RSNA Screening Mammography Breast Cancer Detection AI Challenge, the median recall rate, sensitivity, and specificity were 1.7%, 27.6%, and 98.7%, respectively.
- Combining the top 3 or top 10 algorithms increased the sensitivity to 60.7% and 67.8% and recall rates to 2.4% and 3.5% while decreasing specificity to 98.8% and 97.8%, respectively.
- The sensitivity of the top 3 ensemble model was greater for invasive than for noninvasive cancers (68.0% vs 43.8%; $P = .001$).

Materials and Methods

2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge

In 2023, RSNA invited the scientific and industry community to develop image analysis algorithms that could detect cancer on two-dimensional digital mammograms for breast cancer screening

in the RSNA Screening Mammography Breast Cancer Detection AI Challenge (hereafter referred to simply as the Challenge) (19). The competition host, RSNA, is a nonprofit organization that represents professionals in more than 150 countries. The Challenge was open for submissions from November 2022 until February 2023, with results publicly announced in May 2023.

Evaluation Dataset

The evaluation dataset consisted of 5415 breast screening cases (10830 single-breast examinations), sourced from two sites: site 1, Emory University (United States), and site 2, Breast-Screen Victoria (Australia). Cancer and benign lesions were confirmed with pathologic examination, and no previous or future breast cancers were reported for noncancer cases, with a minimum of 1 year of negative follow-up after screening. No interval cancer cases were included; all cancers were identified at screening. For each case, the dataset included information on patient age, cancer laterality, invasive status (for cancers), and whether a biopsy was performed. Where available, breast density (per Breast Imaging Reporting and Data System, Fifth Edition) was included (20). The manufacturer of the machine used to acquire each examination was either provided by the site or retrieved from Digital Imaging and Communications in Medicine metadata. Example mammograms from the evaluation dataset are shown in Figures 1 and 2. Institutional review board approval was attained at both sites, and data sharing agreements were signed between Challenge organizers and the

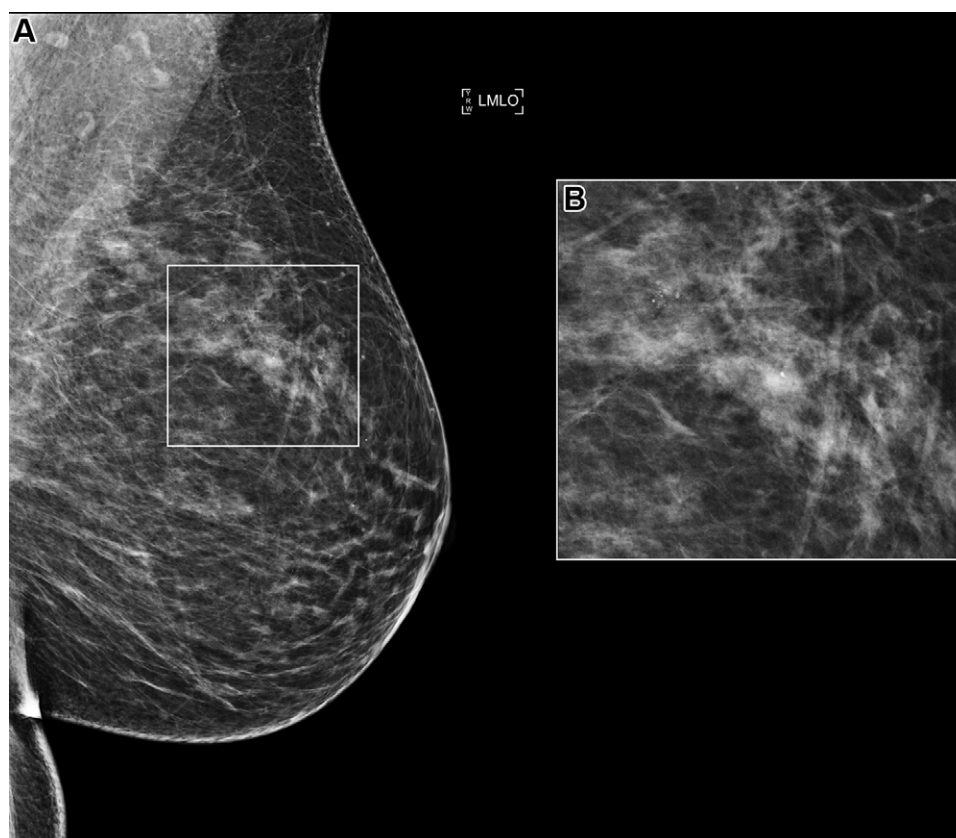


Figure 1: (A) Left mediolateral oblique (LMLO) mammogram in a 58-year-old woman with an area of microcalcification (box). **(B)** Magnified view (2.2×) of the box in **A**. This case was recalled by all of the top 10 artificial intelligence algorithms but was found to be benign at biopsy analysis. This example from the evaluation dataset is from the U.S. site and was acquired using Hologic equipment.

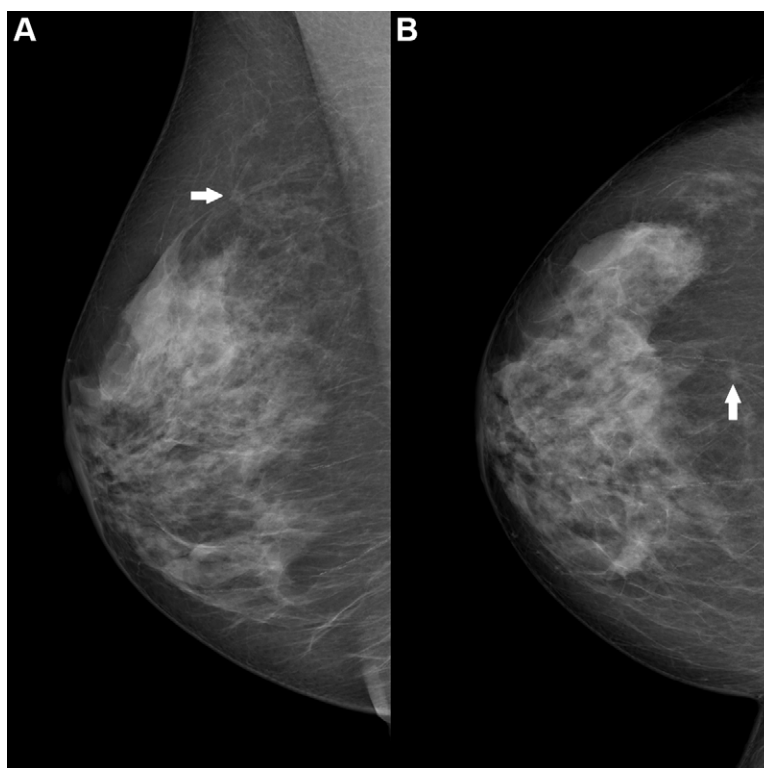


Figure 2: Right breast mammogram in a 69-year-old woman. There is a 6-mm spiculate mass in the 12 o'clock position (arrow), visible on both the (A) mediolateral oblique and (B) craniocaudal views. This case was not recalled by any of the top 10 artificial intelligence algorithms but was a biopsy-proven invasive carcinoma. This example from the evaluation dataset is from the Australian site and was acquired using Siemens Healthineers equipment.

sites. Further details about the Challenge dataset are provided in Appendix S1 and Fig S1.

AI Algorithms

Each team was free to use any architecture for AI image analysis, but since algorithms were run on the Kaggle server for assessment, they were limited by the technical specifications of Kaggle (21). The Challenge provided participants access to a training dataset for algorithm training and testing, consisting of 11 913 screening cases from the same sources as the evaluation dataset (sites 1 and 2)—note that this dataset was distinct from and did not overlap with the evaluation dataset. Teams could also use additional publicly available training data. AI algorithms submitted in the Challenge provided breast-level prediction data, but the format varied: Some algorithms provided binary predictions, whereas others provided a score on a continuous probability scale without a predefined recall threshold. Both approaches were acceptable for the Challenge because performance rankings were based on the probabilistic F1 score (22). For the algorithms providing continuous probability scales, a recall threshold was determined post hoc by selecting the cutoff point that yielded a recall rate equal to the median of the group of algorithms that provided a binary output.

Statistical Analysis

The performance of the AI algorithms was evaluated by computing metrics at the breast level. Algorithms were ranked based on the probabilistic F1 score (22). The probabilistic F1 score weighs the positive predictive value (PPV) of recalling a woman for further assessment following screening against the sensitivity. Because

it optimizes the harmonic mean between these factors, the F1 score is highest when PPV and sensitivity are similar, thus leading to very high specificity. To assess the clinical performance of the algorithms, the following metrics were calculated based on the number of true-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) findings: recall rate ($(TP + FP)/\text{all examinations}$), cancer detection rate ($TP/\text{all examinations}$), PPV ($TP/(TP + FP)$), negative predictive value ($TN/(TN + FN)$), sensitivity ($TP/(TP + FN)$), and specificity ($TN/(TN + FP)$). The distributions of these metrics were computed for all submitted algorithms, with median values and IQRs reported. These metrics were also calculated for ensemble models of the top 3 and top 10 algorithms. Metrics for the ensemble models were computed by recalling a case if any of the included algorithms indicated recall. Binomial CIs were calculated for reported proportion measures.

The sensitivity and specificity of the top-performing AI algorithm and the ensemble models were analyzed for different patient subgroups on the basis of age, breast density, mammogram acquisition site, equipment manufacturer, and cancer type (invasive or noninvasive). Odds ratios (ORs) were used to compare AI performance between subgroups ($\alpha = .05$). For example, for sensitivity, the odds of an algorithm detecting a malignancy among invasive cancers was calculated as the number of true positives among invasive cancers divided by the number of false negatives among invasive cancers, which was then compared with the odds of the algorithm detecting a malignancy among noninvasive cancers (calculated in the same manner).

Analyses were performed using Python version 3.10 (Python Software Foundation).

Table 1: Characteristics of the Evaluation Cohort in the RSNA Screening Mammography Breast Cancer Detection AI Challenge

Characteristic	All Cases	Site 1 (United States)	Site 2 (Australia)
No. of patients	5415	2607	2808
No. of breasts	10 830	5214	5616
Age (y)*	59 (52–66)	57 (48–66)	60 (54–66)
Breast-level reference standard			
Noncancer [†]	10 616 (98.0)	5116 (98.1)	5500 (97.9)
Biopsied	560 (5.2)	403 (7.7)	157 (2.8)
Cancer	214 (2.0)	98 (1.9)	116 (2.1)
Invasive cancer	150 (1.4)	62 (1.2)	88 (1.6)
Benign	346 (3.2)	305 (5.8)	41 (0.7)
BI-RADS density			
A	568 (5.2)	568 (10.9)	NA
B	2166 (20.0)	2166 (41.5)	NA
C	2182 (20.1)	2182 (41.8)	NA
D	284 (2.6)	284 (5.4)	NA
Missing	5630 (52.0)	14 (0.3)	5616 (100)
Equipment manufacturer			
Hologic	6302 (58.2)	4336 (83.2)	1966 (35.0)
Siemens Healthineers	1862 (17.2)	0 (0.0)	1862 (33.2)
Philips	1788 (16.5)	0 (0.0)	1788 (31.8)
GE HealthCare	682 (6.3)	682 (13.1)	0 (0.0)
Fujifilm	196 (1.8)	196 (3.8)	0 (0.0)
Missing	0 (0.0)	0 (0.0)	0 (0.0)

Note.—Except where indicated, data are numbers of breasts, with percentages in parentheses. BI-RADS = Breast Imaging Reporting and Data System, Fifth Edition, NA = not available.

* Data are medians with IQRs in parentheses.

[†] Minimum of 1 year of negative follow-up after screening.

Results

Patient Data

All 5415 cases (10830 single-breast examinations) in the evaluation cohort were screening mammograms, and the median age of the women at screening was 59 years (IQR, 52–66 years) (Table 1). Of the 10830 breasts, 214 (2.0%) had biopsy proven cancer, and 10616 were noncancer cases (98.0%). Of the noncancer cases, 346 breasts (3.2% of total dataset) were biopsied and found to be benign. With respect to Breast Imaging Reporting and Data System density category, 568 (5.2%) breasts were A, 2166 (20.0%) were B, 2182 (20.1%) were C, and 284 (2.6%) were D; breast density was not available for 5630 (52.0%) breasts. Most images were acquired using Hologic equipment (58.2%), followed by equipment from Siemens Healthineers (17.2%), Philips (16.5%), GE HealthCare (6.3%), and Fujifilm (1.8%).

Challenge Submissions

A total of 1687 AI algorithms by 1687 different teams were submitted in the Challenge. Among these, 150 were excluded from analysis because the predictions were not valid (all cases received the same malignancy probability or this value was missing).

A total of 1537 AI algorithms were evaluated: 1072 algorithms provided binary predictions for each case, and 465 algorithms provided a predicted probability of cancer for each case without a predefined recall threshold. For the latter group of algorithms, a recall threshold was determined post hoc by selecting the cutoff point that yielded a recall rate equal to the

median recall rate of the group of 1072 AI algorithms that had a predefined recall threshold.

The algorithms were ranked using the probabilistic F1 metric, with the top algorithm achieving a score of 0.555 (Fig S2). Details of the top 10 algorithms, as ranked by probabilistic F1 score, are provided in Tables 2 and 3.

Recall and Cancer Detection Rate

The distributions of the recall and cancer detection rates among the 1537 AI algorithms evaluated in this study are shown in Figure 3. The median recall rate of the AI algorithms was 1.7% (IQR, 1.4%–2.1%), and the median cancer detection rate was 0.54% (IQR, 0.28%–0.79%) (Table 4). In comparison, the highest-ranked AI algorithm in the Challenge had a recall rate of 1.5% (95% CI: 1.3, 1.7) and a cancer detection rate of 1.0% (95% CI: 0.8, 1.2). The top 3 and top 10 ensemble models had a recall rate of 2.4% (95% CI: 2.2, 2.7) and 3.5% (95% CI: 3.2, 3.9) and a cancer detection rate of 1.2% (95% CI: 1.0, 1.4) and 1.3% (95% CI: 1.1, 1.6), respectively.

Positive and Negative Predictive Value

The distributions of PPV and negative predictive value among the 1537 AI algorithms evaluated in this study are shown in Figure 4. The median PPV among the AI algorithms was 36.9% (IQR, 11.4%–42.2%), and the median negative predictive value was 98.5% (IQR, 98.2%–98.8%) (Table 4). In comparison, the top-ranked AI algorithm and the top 3 and 10 ensemble models achieved PPVs of 64.6% (95% CI: 56.9, 71.6), 49.2% (95%

Table 2: Overview of the Training Methods of the Top 10 Algorithms in the RSNA Screening Mammography Breast Cancer Detection AI Challenge

Rank	Training Method					GitHub Link*
	External Datasets Used	Preprocessing	Data Augmentation	Multistep Training	Auxiliary Task	
1	BMCD, CDD-CESM, CMMD, Mini-DDSM, VinDr-Mammo	YOLOX, crop	Brightness and contrast adjustment, coarse dropout, downscale, elastic transform, grid distortion, random resized crop, shift-scale-rotate, tone curve adjustment, vertical and horizontal flip	No	No	https://github.com/dangnh0611/kaggle_rsna_breast_cancer
2	DDSM-positive cases	Faster R-CNN, crop	Random augment, random flip, random erasing, shift-scale-rotate	Yes	Yes	https://github.com/ShuzhiLiu/RSNABreast2ndPlace
3	CBIS-DDSM, CMMD, Mini-DDSM, VinDr	YOLOX, crop	Brightness and contrast adjustment, coarse dropout, elastic transform, Gaussian blur, Gaussian noise, grid distortion, median blur, mixup, optical distortion, shift-scale-rotate, vertical and horizontal flip	Yes	No	NA
4	DDSM, VinDr-Mammo	Coarse CNN with minimum filter to crop image to breast area, crop	Affine transform, coarse dropout, grid distortion, grid shuffle, random crop, shift-scale-rotate, transpose, vertical and horizontal flip	Yes	Yes	NA
5	DDSM, VinDr-Mammo	YOLOX, crop	Brightness and contrast adjustment, coarse dropout, shift-scale-rotate (breast regions), vertical and horizontal flip	Yes	No	https://github.com/nhannguyen2709/rsna-breast
6	None	YOLOX, crop	Affine transform, blur, brightness and contrast adjustment, contrast limited adaptive histogram equalization, distortion, dropout, vertical and horizontal flip	Yes	No	https://github.com/tyamaguchi17/rsna_mammo
7	VinDr-Mammo	Crop	Affine transform, elastic transform, vertical and horizontal flip	Yes	No	https://github.com/Masaaaato/RSNABreast7thPlace
8	VinDr-Mammo	Crop	Affine transform, brightness and contrast adjustment, JPEG compression, light and heavy augmentations, random resized crop, shift-scale-rotate, vertical and horizontal flip	Yes	No	https://github.com/chqwer2/RSNA_Solutions/tree/main/RSNA_2023_8thSolution_Screening_Mammography_Breast_Cancer_Detection
9	None	Crop	Affine transform, brightness and contrast adjustment, coarse dropout, gamma adjustment, grid dropout, rotate, shift-scale-rotate, vertical and horizontal flip	No	No	NA
10	None	Crop	Brightness and contrast adjustment, coarse dropout, shift-scale-rotate, vertical and horizontal flip	Yes	Yes	https://github.com/moritake04/rsna-breast-cancer-detection

Note.—BMCD = Breast Micro-Calcifications Dataset, CBIS-DDSM = Curated Breast Imaging Subset of DDSM, CMMD = Chinese Mammography Dataset, CNN = convolutional neural network, DDSM = Digital Database for Screening Mammography, NA = not available, R-CNN = Region-based CNN.

* Discussions and descriptive readme files for the top 10 algorithms and other teams can be accessed via the Kaggle leaderboard for the challenge: <https://www.kaggle.com/competitions/rsna-breast-cancer-detection/leaderboard>.

Table 3: Overview of the Model Designs of the Top 10 Algorithms in the RSNA Screening Mammography Breast Cancer Detection AI Challenge

Rank	Model Design					Model Reference	
	Input Size	Input View Level	Backbone Model	Modules	No. of Ensemble Models	Total Reference Speed (sec)*	Test Time Augmentation
1	2048 × 1024	Single view	ConvNeXt-v1-Small	None	4	28 418.4	None
2	1280 × 1280, 1536 × 1536	Single view, breast, exam	Convnext-v1-Small	None	1	20 104.5	Diagonal flip
3	1536 × 960	Multiview	EfficientNet-v2-Small, ConvNeXt-Nano, ConvNeXt-Tiny	Long short-term memory	11	29 428.7	None
4	1024 × 1024	Single view, breast, exam	EfficientNet-v2-Small, EfficientNet-v2-Medium, EfficientNet-b3, EfficientNet-b4, EfficientNet-b5, SE-ResNeXt50, ConvNeXt-Tiny	Multiview-multilateral transformer	7	20 147.9	None
5	1536 × 1024	Single view	EfficientNet-v2-Small, EfficientNet-b5-Small	None	10	18 045.8	None
6	1024 × 512, 1536 × 768	Multiview, multiview multilaterality	ConvNeXt-Small, ConvNeXt-Nano, ConvNeXt-Base	None	4	28 849.5	Vertical flip
7	1520 × 912	Multilaterality	EfficientNet-v2-Small, EfficientNet-b5	None	4	26 329.5	None
8	1536 × 896	Single view	EfficientNet-v2-small, ConvNeXt-Nano, ECA-NFNet-l0	None	3	NA	Horizontal flip; binarization postprocessing
9	1536 × 768	Single view	ConvNeXt-v1	None	3	NA	Horizontal flip
10	1536 × 960, 1536 × 768	Single view	EfficientNet-v2-Small, MaxViT-Tiny	None	3	NA	Horizontal flip

Note.—NA = not available.

* Including image preprocessing.

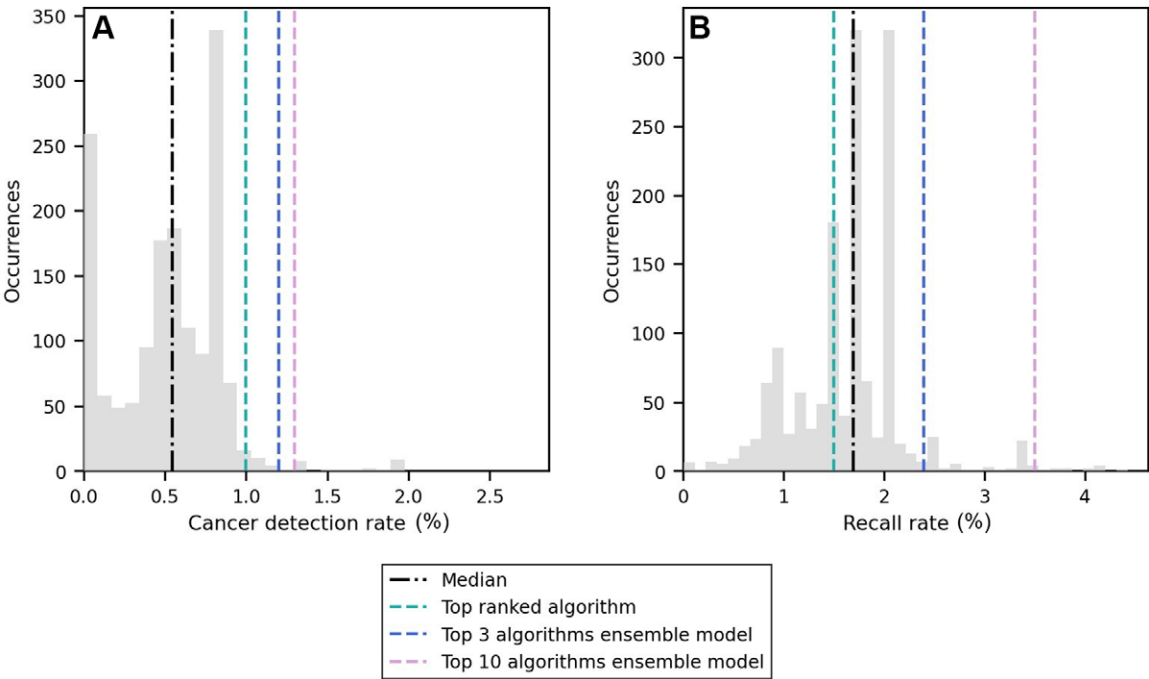


Figure 3: Distribution of **(A)** cancer detection rate and **(B)** recall rate among the 1 537 artificial intelligence algorithms evaluated in this study (gray bars). The median value is represented by the black line, and the scores of the top-ranked algorithm and the two ensemble models are shown with colored lines.

Table 4: Performance Metrics for All AI Algorithms, the Highest-ranked Algorithm, and the Top 3 and Top 10 Ensemble Models

Metric	All 1537 AI Algorithms*	Top AI Algorithm†	Top 3 Ensemble Model†	Top 10 Ensemble Model†
Recall rate (%)	1.7 (1.4–2.1)	1.5 (1.3, 1.7) [161/10830]	2.4 (2.2, 2.7) [264/10830]	3.5 (3.2, 3.9) [383/10830]
CDR (%)	0.54 (0.28–0.79)	1.0 (0.8, 1.2) [104/10830]	1.2 (1.0, 1.4) [130/10830]	1.3 (1.1, 1.6) [145/10830]
PPV (%)	36.9 (11.4–42.2)	64.6 (56.9, 71.6) [104/161]	49.2 (43.3, 55.2) [130/264]	37.9 (33.1, 42.8) [145/383]
NPV (%)	98.5 (98.2–98.8)	99.0 (98.8, 99.1) [10561/10669]	99.2 (99.0, 99.4) [10484/10566]	99.4 (99.2, 99.5) [10380/10447]
Sensitivity (%)	27.6 (14.0–40.2)	48.6 (42.0, 55.3) [104/214]	60.7 (54.1, 67.0) [130/214]	67.8 (61.2, 73.7) [145/214]
Specificity (%)	98.7 (98.3–99.2)	99.5 (99.3, 99.6) [10561/10616]	98.8 (98.5, 98.9) [10484/10616]	97.8 (97.5, 98.0) [10380/10616]
Probabilistic F1 score	0.306 (0.080–0.392)	0.555 (0.463, 0.643)	0.544 (0.451, 0.634)	0.486 (0.393, 0.580)

Note.—Top-ranked AI algorithm and top 3 and top 10 ensemble models were based on ranking in the 2023 RSNA Screening Mammography Breast Cancer Detection AI Challenge, using the probabilistic F1 score. For the ensemble models, a case was recalled if any of the constituent algorithms recalled the case. AI = artificial intelligence, CDR = cancer detection rate, NPV = negative predictive value, PPV = positive predictive value.

* Data are medians, with IQRs in parentheses.

† Data are percentages, with binomial CIs in parentheses and numbers of breasts in brackets.

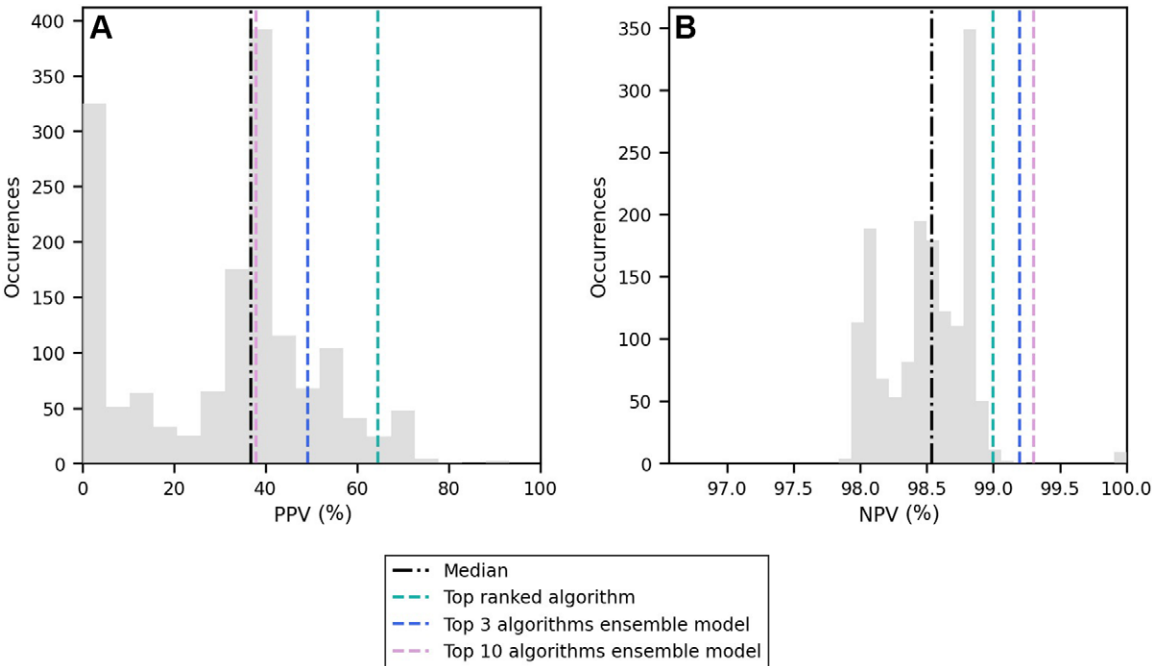


Figure 4: Distribution of (A) positive predictive value (PPV) and (B) negative predictive value (NPV) among the 1537 artificial intelligence algorithms evaluated in this study (gray bars). The median value is represented by the black line, and the scores of the top-ranked algorithm and the two ensemble models are shown with colored lines.

CI: 43.3, 55.2), and 37.9% (95% CI: 33.1, 42.8) and negative predictive values of 99.0% (95% CI: 98.8, 99.1), 99.2% (95% CI: 99.0, 99.4), and 99.4% (95% CI: 99.2, 99.5), respectively.

Sensitivity and Specificity

The distributions of sensitivity and specificity among the 1537 AI algorithms evaluated in this study are shown in Figure 5. The median sensitivity of the AI algorithms was 27.6% (IQR, 14.0%–40.2%), and the median specificity was 98.7% (IQR, 98.3%–99.2%) (Table 4). The highest-ranked AI algorithm achieved a sensitivity of 48.6% (95% CI: 42.0, 55.3) and a specificity of 99.5% (95% CI: 99.3, 99.6). The top 3 and top 10 ensemble models had a sensitivity of 60.7% (95% CI: 54.1, 67.0)

and 67.8% (95% CI: 61.2, 73.7) and a specificity of 98.8% (95% CI: 98.5, 98.9) and 97.8% (95% CI: 97.5, 98.0), respectively.

Influence of Patient Characteristics on Performance

The sensitivity and specificity of the highest-ranked AI algorithm and the ensemble models as a function of different patient characteristics are shown in Figure 6 (more details can be found in Table S1). Sensitivity differed across subgroups stratified by cancer type, acquisition site, and equipment manufacturer. Sensitivity was significantly greater for invasive cancer cases than for noninvasive cancer cases for the top-ranked algorithm (56.7% vs 29.7%; OR = 3.10; $P < .001$), the top 3 ensemble model (68.0% vs 43.8%; OR = 2.73; $P = .001$), and the top 10 ensemble model

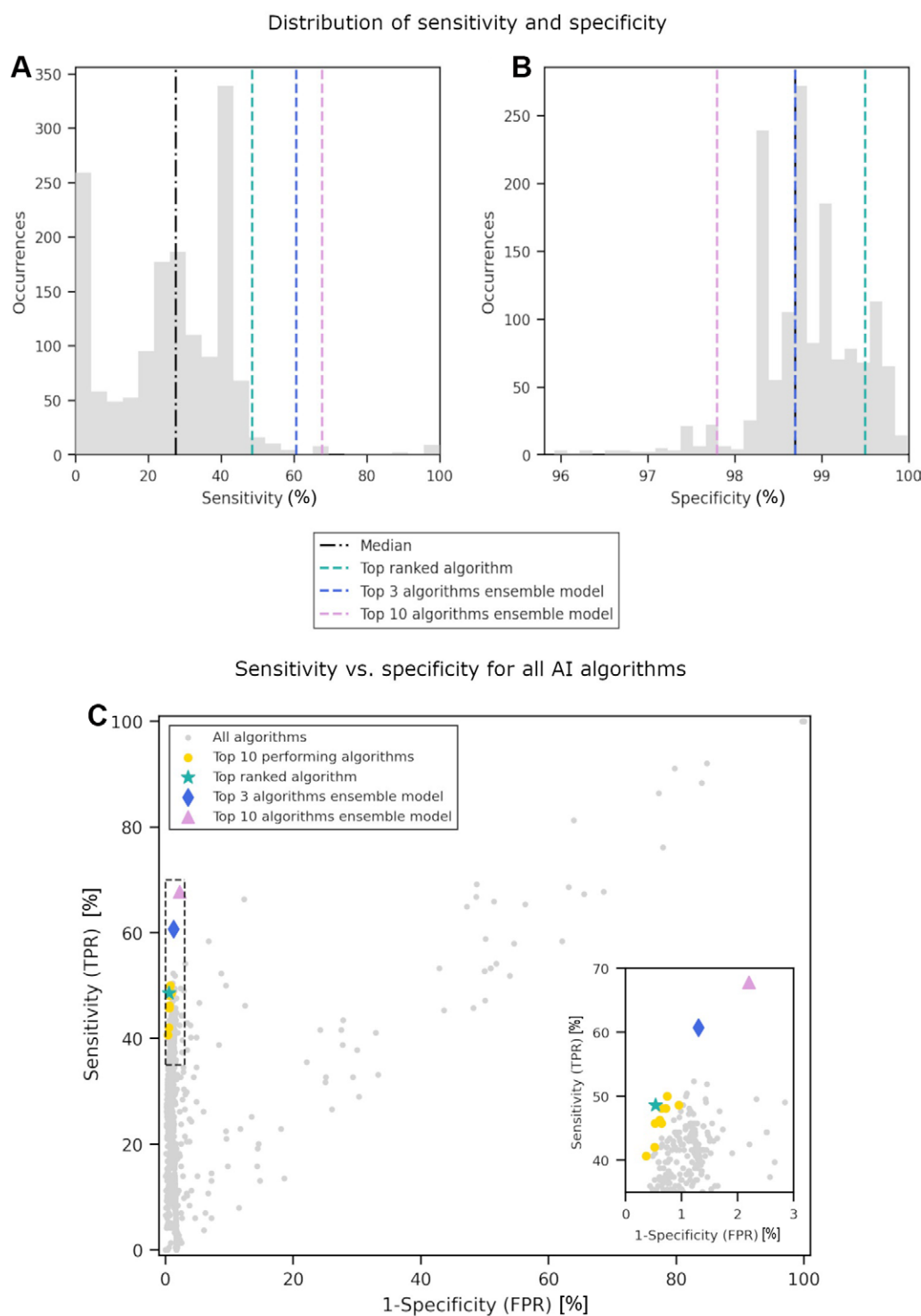


Figure 5: Distribution of (A) sensitivity and (B) specificity among the 1537 artificial intelligence (AI) algorithms evaluated in this study (gray bars). The median value is represented by the black line, and the scores of the top-ranked algorithm and the two ensemble models are shown with colored lines. (C) Scatterplot of sensitivity versus specificity for each AI algorithm, where each gray dot represents a single AI algorithm. The inset on the right provides an expanded view of the area in the dashed box, so that the distribution of the top-performing algorithms can be seen at greater resolution. FPR = false-positive rate, TPR = true-positive rate.

(74.7% vs 51.6%; OR = 2.77; $P = .001$). Sensitivity was significantly lower for site 1 (United States) than for site 2 (Australia) for the top-ranked algorithm (40.8% vs 55.2%; OR = 0.56; $P = .04$) and the top 3 ensemble model (52.0% vs 68.1%; OR = 0.51; $P = .02$). Finally, sensitivity was lower for mammograms acquired using Hologic equipment compared with equipment from other vendors for the top 3 ensemble model (54.4% vs 71.8%;

OR = 0.47; $P = .01$) and the top 10 ensemble model (62.5% vs 76.9%; OR = 0.5; $P = .03$). Sensitivity was similar when comparing cases grouped according to patient age or breast density.

Specificity differed significantly for cases grouped according to age, site, or equipment manufacturer. Specificity was greater for patients younger than 59 years than for patients aged 59 years or older for the top-ranked algorithm (99.6% vs 99.3%;

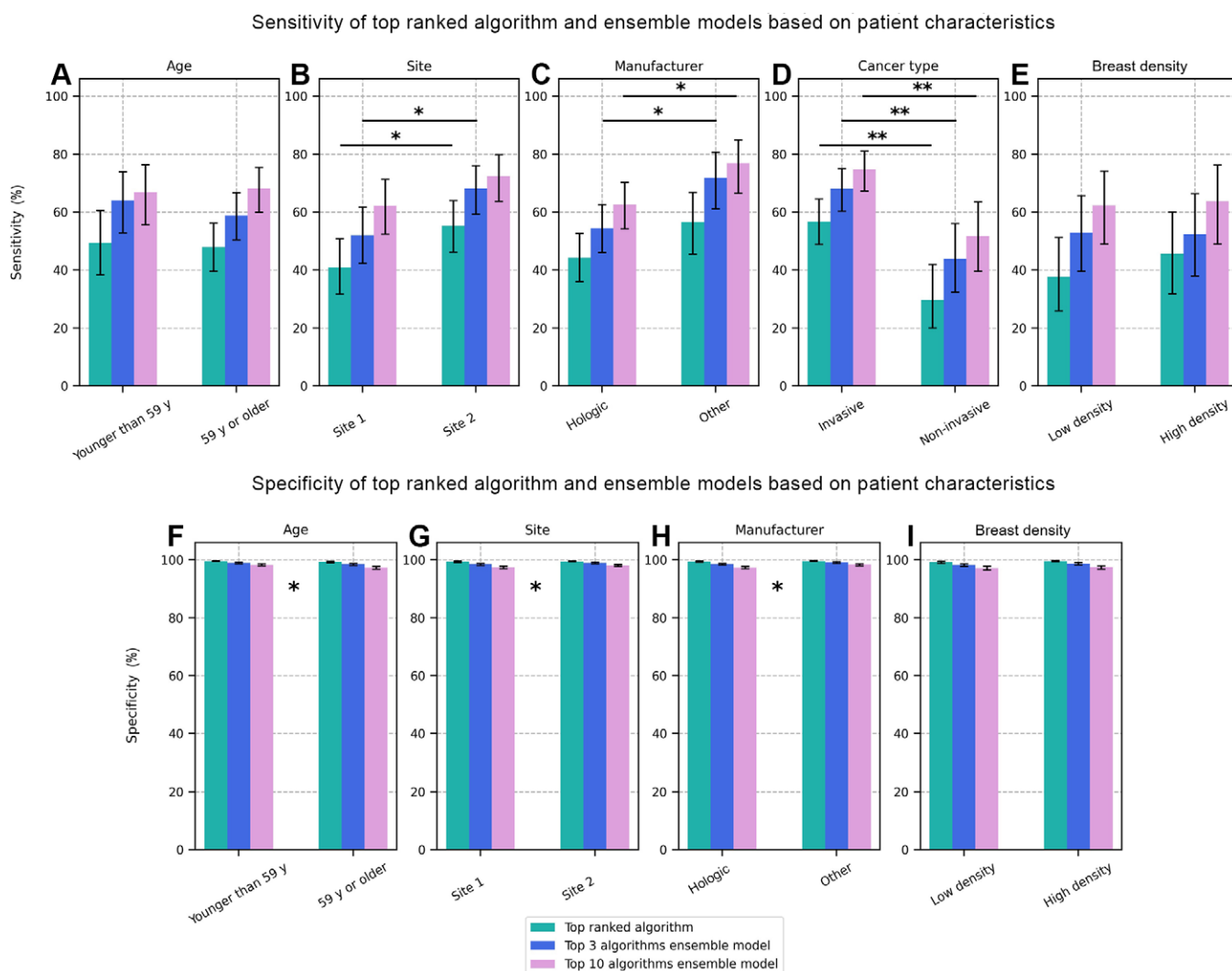


Figure 6: Bar plots showing the (A–E) sensitivity and (F–I) specificity of cancer detection by the top-performing artificial intelligence (AI) algorithm and the top 3 and top 10 ensemble models when the dataset was stratified as a function of different patient characteristics: (A, F) age, (B, G) site 1 (United States) versus site 2 (Australia), (C, H) equipment manufacturer (Hologic vs other [GE Healthcare, Fujifilm, Philips, or Siemens Healthineers]), (D) invasive versus noninvasive cancer, and (E, I) low-density breasts (Breast Imaging Reporting and Data System A or B) versus high-density breasts (Breast Imaging Reporting and Data System C or D). For the sensitivity plots, significant differences between subgroups are indicated: * = $P < .05$, ** = $P < .005$. Significant differences between subgroups are not indicated for specificity plots because of the relatively small differences in magnitude, but * indicates characteristics for which there was a significant difference between subgroups for one or more of the AI entities.

OR = 1.99; $P = .02$) and the top 10 ensemble model (98.3% vs 97.3%; OR = 1.61; $P < .001$). Specificity was lower for site 1 than for site 2 for the top 3 ensemble model (98.5% vs 99.0%; OR = 0.69; $P = .03$) and the top 10 ensemble model (97.4% vs 98.1%; OR = 0.70; $P = .008$). Specificity was also lower for mammograms acquired with Hologic equipment versus equipment from other vendors for the top 3 ensemble model (98.5% vs 99.1%; OR = 0.56; $P = .003$) and the top 10 ensemble model (97.3% vs 98.5%; OR = 0.59; $P < .001$). There was no evidence of a difference in specificity for low-density versus high-density breasts.

Discussion

Here, we report the performance of algorithms for breast cancer detection at mammography submitted to the RSNA Screening Mammography Breast Cancer Detection AI Challenge. Participation was strong, with 1687 different algorithms submitted. Of these, 1537 algorithms produced meaningful classification output. The algorithms were ranked using the probabilistic F1 score, with the top algorithm achieving a score of 0.555. In our

study, performance was translated into more clinically useful metrics, including recall rate, sensitivity, and specificity. The top algorithm yielded a recall rate of 1.5% in the evaluation dataset and achieved a sensitivity of 48.6% and a specificity of 99.5%. Ensemble models, combining the predictions of multiple high-ranking AI algorithms, had increased sensitivity (compared with the top algorithm): 60.7% and 67.8% for ensemble models of the top 3 and top 10 AI algorithms, respectively, with corresponding recall rates increased to 2.4% and 3.5% and specificities decreased to 98.8% and 97.8%. Sensitivity was influenced most by acquisition site and mammography machine vendor, with lower sensitivities observed for site 1 (United States) than for site 2 (Australia) and for Hologic equipment compared with equipment from other vendors.

While the F1 scores achieved by the algorithms were high, the clinical performance of the algorithms appeared lower. All cancers were screen detected and therefore visible to human observers; the maximum sensitivity for a single algorithm was 48.6%, which is relatively low. However, the exclusion of interval cancers and

cancers detected in subsequent screening rounds from the Challenge dataset artificially increased the radiologists' sensitivity to 100%, and potential true positives identified by AI algorithms but not detected by radiologists at screening were disregarded. In other words, the bias favoring the radiologists was maximized. Therefore, it is logical that even very good algorithms did not achieve the same sensitivity as radiologists.

A further factor that explains the relatively low sensitivity of the top-performing algorithms is the very low recall rate enforced by the ranking method (probabilistic F1 score). This low recall rate resulted in the impressively high PPV of 64.6% for the top-performing algorithm. In clinical practice, however, a higher false-positive rate is accepted to optimize sensitivity; in general, a PPV of approximately 20% is deemed acceptable (23). In fact, a recent meta-analysis that evaluated mostly commercial AI algorithms tested on large retrospective cohorts showed that widely published algorithms were generally less specific than radiologists reading the same studies; on average, sensitivity was 75.8% for algorithms compared with 72.6% for radiologists, at a specificity of 95.6% for the algorithms compared with 96.4% for the radiologists (24).

With this in mind, we created ensemble models to increase sensitivity at the cost of specificity. The evident increase in performance shows that various AI algorithms can provide complementary information and may strengthen each other's results. Use of the ensemble model that consisted of the 10 highest-ranked algorithms increased the sensitivity from 48.6% (for the top-ranked algorithm) to 67.8% while increasing the recall rate from 1.5% to 3.5%. This higher recall rate is similar to that observed in many European screening programs and, as such, may provide a more realistic target value for the AI algorithms submitted in the Challenge. In a study from Sweden, high-volume screen readers had a sensitivity that ranged from 63% for the lowest quartile to 84% for the highest quartile, with specificities ranging from 95% to 98% (25). The performance of the top 10 ensemble model from the Challenge seemed to be slightly below the average performance of these high-volume readers but above the average performance of the lowest quartile of high-volume readers.

A striking finding is that the performance of the AI algorithms in the cohort from site 1 (United States) was worse than that in the cohort from site 2 (Australia). This finding may be explained in part by differences in population and screening equipment. However, differences in recall policy and workup between the countries likely also contribute to this finding. Recall rates for follow-up evaluations are approximately 10% in the United States but less than 4% in Australia. The higher recall rate in the United States allows workup for much more subtle signs of cancer; thus, cancers are detected that would remain unnoticed in the Australian population, at the cost of a substantial drop in specificity. It is therefore possible that cancers from site 2 were simply somewhat easier to detect than those from site 1. Consequently, even the ensemble models are less applicable to the data from site 1. The performance of the top 10 ensemble model in the site 2 cohort is likely a realistic prediction of clinical performance in European and Australian settings, but for a United States–based screening program, either the ensemble should be expanded or algorithm thresholds should be adapted to fit with clinical reality.

The lower performance of the algorithms on mammograms obtained with machines from Hologic is most likely a result of

the same issue. At site 1, 83.2% of mammograms were obtained with machines from Hologic, whereas at site 2, 65.0% were obtained with machines from Philips or Siemens Healthineers. Furthermore, breast density did not affect the performance of the algorithms. This finding may appear counterintuitive but follows from the fact that all positive cases were screen detected and therefore visible to human readers.

One limitation of the study is that many teams designed algorithms (including the top 10) that provided a binary score (cancer present or absent) for each breast rather than a score on a continuous probability scale, as seen in commercial algorithms, which limited our ability to tune algorithm performance by manipulating the recall threshold. A second limitation is that we did not evaluate how algorithmic architecture and integrated AI backbone networks affected algorithm performance. Therefore, we cannot address the optimal way to design AI algorithms for mammographic screening. However, the solutions of the leading teams, as well as several others, are available via the Kaggle leaderboard for the Challenge (<https://www.kaggle.com/competitions/rsna-breast-cancer-detection/leaderboard>) and are fully open-source. A third limitation is that the performance of the AI algorithms was validated on a test set obtained from the same data sources as the training dataset. While this is standard in machine learning, it prohibits assessment of the generalizability of the algorithms to out-of-distribution populations, such as populations in other countries that may differ in cancer rates, screening regimens, or equipment.

In conclusion, we report the results of what is to our knowledge the largest challenge for mammographic screening thus far performed. Our study revealed that different artificial intelligence (AI) algorithms found different cancers on screening mammograms and could be used in an ensemble to increase sensitivity while maintaining clinically realistic recall rates. An ensemble of the 10 best-performing algorithms yielded performance close to that of an average screening radiologist in Europe or Australia. The algorithms could be used to further optimize existing AI applications for breast cancer detection on screening mammograms. We hope that the results of the Challenge may contribute to the further improvement of both experimental and commercial AI tools for mammography, so that all women can benefit from the results of this challenge.

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